

Zero-shot Generalization of Multimodal Dialogue Agents

Diogo Tavares
Universidade NOVA de Lisboa, NOVA LINCS
Lisbon, Portugal
dc.tavares@campus.fct.unl.pt

ABSTRACT

Multimodal conversational agents are an ever expanding field which benefits from the introduction of large language models. Production-ready robust conversational assistants trade breadth of scope for higher accuracy and general dialogue quality. These conversational assistants must be able to maintain the conversation focused, respond appropriately to user requests, maintain a certain level of natural response generation, be robust to out-of-scope and chitchat attempts, and, of course, be accurate in assisting the user in reaching their domain-specific goals. This work discusses data-centric observations, alongside providing research hypothesis for future, and some of my already developed work, to be expanded throughout my PhD.

CCS CONCEPTS

• Computing methodologies → Natural language processing.

KEYWORDS

Dialogue Agents, Natural Language Processing, Natural Language Understanding, Zero-shot, Slot-filling, Dialogue State Tracking, Intent Detection

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1 INTRODUCTION

In recent years, we've seen the advent of very large dialogue-focused language models, such as BlenderBot [20] and LaMDA [4]. These generally outperform smaller models in open-ended conversation, but are impractical to fine-tune and sometimes even to perform inference on. Furthermore, their sizes are said to be overblown for the amount of tokens they tend to be trained on [3]. Even when used in zero-shot settings, it's mostly on the *task* level, and not on the *domain* level, meaning performance is still relatively low [23]. The conversational system developer, to approach that level of performance using smaller models, must understand and

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User: I'm looking to buy a t-shirt.

Assistant: What kind of t-shirt are you looking for?

U: Hmm... Something like this but with blue patterns.



A: Okay, let me look for something in my catalogue... Here you go!



U: Is polyester machine washable?

A: Yes; note that you should use warm water.



U: Can you also tell me if **eat your peas, son .. it'll rain tomorrow?**

A: That's highly dependant on your specific location. Check your local weather forecast!



U: Ok, add the t-shirt with the chest pocket to my cart. Bye bye.

A: Sure! Adding it to your cart. Have a wonderful day.

Figure 1: Main research challenges addressed in this doctoral plan. We highlight them as follows: 1- Cross-modal knowledge; 2- Real world zero-shot knowledge for QA; 3- Handling out-of-scope and noisy utterances; 4- Dynamically accessing real-world APIs

adapt them to their specific use case, by transferring domains. Narrowing the scope to focus on single-domain applications (such as hotel booking systems, cooking companions or music recommenders) may maximize correctness and improve overall functionality. Other approaches may include inserting multimodal aspects to the system, which can unlock several novel and interesting interactions.

In this work, I focus on the topic of Natural Language Understanding in narrow-scoped conversational assistants, and discuss some of the insight I've gained while participating in the **Alexa Prize Taskbot Challenge** [12] (henceforth, the *competition*) as a member of the Twiz team [7]; and developing a **Multimodal Conversational Agent for Fashion**. More specifically, I'll discuss why I think NLU is important (section 1.2), risks for its viability in production environments (sections 1.3 and 2, which is specifically data-related), and research topics I'd like to see addressed—by me—in the future (sections 3 and 4).

1.1 Dialogues, Slot-filling, Intent detection

Typical task-oriented conversations differ from chit-chat dialogues in a few key areas. Properly understanding natural (and spoken) language, which is inherently noisy and complex, is paramount when aiming to successfully complete a task. A *hotel reservation* system needs to **know** the location a user intends to travel to. A

chit-chat agent, on the other hand, doesn't need to **remember** that the user's favorite fruit is the apple—though it would be a positive. These long-term dependencies need to be maintained either implicitly—in some *dialogue state*—or explicitly, by connecting relevant domain concepts (**slots**) to their respective **values**, as per user preference. Explicitly remembering what a user wants comes with some clear benefits: high explainability to the developer, the maximization of recall in later sections of the dialogue, and, potentially, an improvement on the interaction quality: the system can smoothly repeat the user's wants back to them in a natural way. Explicitly remembering information is also natural to humans: when agents perform **slot-filling**, they are merely making a *mental note* of the user's preferences, which would be reasonable to expect from a human. Humans, however, don't perform explicit **intent detection**. Classifying user utterances into a restricted set is a concession that comes with clear advantages: anything a user says will always be mapped¹ to the best possible dialogue state. The intent detector can also classify whether an utterance is in or out-of the system's scope—robust, real-world systems, such as the ones developed for the *challenge*, have much to gain from smoothly handling out-of-scope utterances. In typical explicit intent detection, intents are thus mapped into a **developer-defined restricted set of possible values**. Although this set is restricted by definition, it must attempt to be general enough to capture as many user interactions as possible. Later on, in figure 4, we'll show the intents we, as dialogue system developers, defined when participating in the *competition*.

A pitfall of this explicit classification is that restricting all user utterances into a limited set doesn't necessarily translate to an ever evolving system: adding and removing functionality must update the set of possible intents. This leads to two interesting **challenges**: **1. how to introduce an intent detector module where intents can be dynamically updated** and **2. integrate such a model with other elements of the dialogue pipeline which are generally independent from the NLU section**, such as a dialogue manager or a response generator, **without heavily relying on handcrafted rules**.

1.2 NLU directly influences dialogue quality

The importance of NLU lies in its ubiquity in any situations where users interact with a system using voice or text; and its failure impeding the dialogue from progressing satisfyingly. It's especially important because it's where the user input is directly fed; if these models misunderstand the input, the rest of the dialogue pipeline (e.g. some internal reasoner or a response generator) will produce incorrect outputs.

Intent detection failing may catastrophically alter the course of a dialogue: the dialogue manager will fail to properly map the user input to a dialogue state, and the system response will be **wrong**. The user may then react unexpectedly, and the conversation will significantly veer off course. Slot-filling (and State Tracking) failing may lead to a more insidious outcome: wrongly classifying a certain

slot may have implications much later in a dialogue. If a user requests *hotels near A*, but the system classifies *hotel-location* as *B* and doesn't ask for user confirmation, the conversation may proceed to *restaurant* and *taxi* reservations without the user noticing the misclassification. In this case, correcting the system's belief state may set the dialogue back by several utterances.

Systems must mitigate these flaws by failing both tasks as gracefully as possible, having some general understanding of out-of-scope intents, and employing some (un)certain measure when classifying slots—e.g. asking for confirmation when certainty is low. And, of course, generally having high classification performance.

1.3 On generalization and scaling

Currently available NLU data doesn't apply to every possible domain. Approaches to create a novel NLU system (i.e., for the *competition*), will either consist of **iteratively increasing the complexity**, by starting with hand-crafted rules and annotating user data to train the models, or by employing **zero or few-shot approaches** (discussed further in section 3). In non-novel domains, systems tend to be trained on common datasets, such as MultiWOZ [6], SGD [19], SNIPS [5], and CLINC150 [14].

Throughout the *competition*, we employed the aforementioned iterative approach, and stumbled upon an issue which affected us, and I suspect may affect any approach relying on synthetic or crowdworker-based datasets: the clear distinction of interaction patterns between regular users and crowdworker data. This issue has the potential to heavily impact generalization, not from training to test data, but from evaluation metrics to in-the-wild performance.

2 LESSONS FROM THE TRENCHES

Instructed users interact with dialogue systems differently from in-the-wild users. During the competition, team members could easily identify these user classes, namely due to instructed dialogues being composed of longer (figure 2) but less varied dialogues, while in-the-wild users use more natural and off-topic utterances, with a greater number of unique tokens (figure 3). Instructed users are more *exploitative* while in-the-wild users are more *explorative*. To confirm that these distinctions are relevant on a local and global level, we created two variations of a classification dataset: *Dialogue*, where each example contains a whole dialogue, and *Utterance*, where examples are the non-overlapping utterances between both sets. Statistics for both datasets—a subset of all dialogues that happened in the competition for which we had annotations—are shown in table 1. We also employ a BERT-based classifier to distinguish between both types of dialogues and corroborate our suspicions, with classification results also shown in table 1. This disparity in ways both user classes interacted with our taskbot was also felt in intent detection performance, which saw a **10% accuracy drop when facing in-the-wild users**. Figure 4 showcases the intent distributions per user type, taken from the intersection of data for which we have both hand-annotated intent ground truth and user type annotations. Note how instructed users generally don't explore all the available features, and instead focus on completing the task as efficiently as possible, only activating a small subset of supported intents. This could introduce bias into the data used to train the intent detection model. This kind of bias is not necessarily a new

¹by some dialogue manager module, using the intent classification as its main input feature, and mapping it to some function call, oftentimes using some of the classified slots as input arguments

topic [11] and is present in several public datasets, with our work further confirming its prevalence, even in spoken language data. This leads us to another highly relevant **third research challenge: the clear distinction between instructed and in-the-wild user utterances** when it comes to NLU.

The same observations can be made for widely popular conversational agent datasets, such as the synthetic MMD [21]—where utterance variability is notoriously low for each (sub)intent—but also the crowdworker-focused MultiWOZ, where user utterances always heavily veer towards dialogue completion and don’t contain any of the noise typically seen in real world conversations. These clear distinctions increase the risk of lowering the performance of a dialogue system in production. Some large-scale datasets, such as the recently released MASSIVE [8] (an extension of the **crowd-sourced** SLURP [1]), could mitigate this problem—back-translating all utterances to English could even introduce some language artifacts, which would maximize intra-intent variability. These noisy utterances may minimize bias, and improve generalization to in-the-wild user data. Lessening our reliance on the utterances being well-formed and always on-topic, by performing zero or few-shot intent detection, could also improve generalization in the long-term. Nevertheless, I believe tackling this bias is a good way of improving dialogue system performance in the long term. This leads directly into the research hypotheses I present: **generalizing dialogue agents with as little data as possible means data collection of real-world dialogues becomes feasible.**

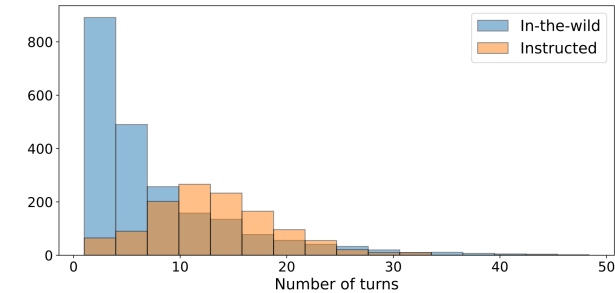


Figure 2: Dialogue length per user type. There is a clear distinction between dialogues by in-the-wild and instructed users.

	Dialogue		Utterance	
	In-the-wild	Instructed	In-the-wild	Instructed
Label count	1766	968	3518	1307
Classification acc.	0.94		0.87	

Table 1: Label count (training set) and classification results (test set) of both datasets

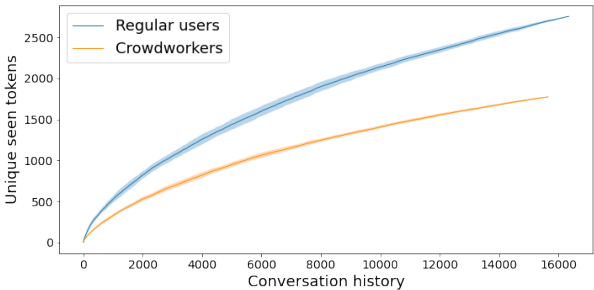


Figure 3: Total unique token counts, per utterance, with the line representing the mean of the values over 20 randomly shuffled iterations and the shaded area representing the standard deviation. Once again, the distinction between user types is clear.

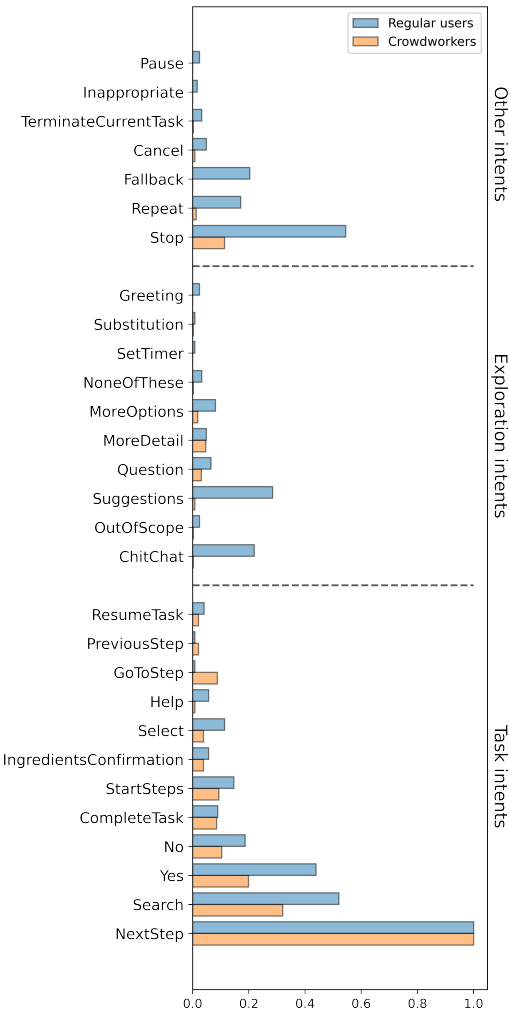


Figure 4: Normalized intent distributions.

Context : I'm looking to make a cherry chocolate cake.

Question: Is the user looking to bake a cake?

Answer : Yes

Question: Is the user asking about the tools they will need?

Answer : No

Question: What kind of cake does the user want to bake?

Answer : cherry chocolate cake

Question: What kind of kitchen does the user have?

Answer : (empty)

Listing 1: Reading comprehension example for NLU. Out-of-scope intents are handled by responding with an empty result

3 RESEARCH HYPOTHESIS 1: ZERO-SHOT INTENT DETECTION AND SLOT-FILLING

Most conversational systems in production will be updated with new features throughout their lifetime. These features may require new intents for which there is no training data. **Heavily instructed (such as crowdsourced) data doesn't necessarily translate well to real-world usage** (section 2), and we presume the same applies to synthetic data—real user interactions are preferred. One then finds oneself in a chicken-or-the-egg situation, where data is required to train the system, but the system needs to support the new feature and associated user interactions in order to acquire said data. I believe zero-shot intent detection and slot-filling is a worthwhile research path, as it can lead to the creation of good-enough solutions to these tasks, which can be good enough to collect data to be leveraged in the long-term.

To tackle this problem, I plan to first assess approaches that pre-processing utterances with keyword-extraction methods: perhaps all a dialogue manager needs is a **verb and a noun—the action and its target**—to perform an appropriate action. Or explicitly **inserting knowledge from external KBs** to perform basic reasoning over user utterances—which may lead to discovering the true intent even when it is not explicitly available as a classification target [22]. As a second approach, I will investigate **domain-transfer approaches from widely used NLU or QA datasets** [18], where data may be similar enough to what the system may require in order to respond successfully—following a reading comprehension approach [9; 10; 16] (listing 1). Generally, semantically classifying the user utterances over some representative embedding of the *intent* can remove the need to use a predefined set [15]. If zero-shot approaches are sufficiently performant, there may not even be a need to retrain the model to support the new intents. A robust zero-shot intent detector can answer **two of the challenges**, by **adding the support for dynamically updating user intents** but also by **generally lessening the need for high quality and plentiful, data**.

4 RESEARCH HYPOTHESIS 2: ZERO-SHOT STATE AND ANSWER GENERATION

The intent of an utterance directly influences the referenced slots and the flow of the conversation. As both of these NLU subtasks go

hand-in-hand, jointly modeling and training them should improve performance [2; 17]—this should happen even when the available intent annotations are very coarse-grained. I believe, then, that the related tasks of dialogue management and response generation should be jointly trained, alongside NLU, following the steps of pretrained language models for task-oriented dialogues [13; 25].

I plan on combining the introduction of zero-shot intents and slot-filling outputs into a representative embedding of the dialogue state. This embedding can then be used either within the same, large-scale, model, or by another model later in the pipeline, which is jointly trained. Explicit values for slots and intents may then be acquired, for example, by the means of prompt engineering, to maximize explainability. I believe it to be intuitive that this dialogue state-aware (instead of intent-aware) model *could* dynamically manage the dialogue and system responses without directly relying on the limited set of intents and slot names, which could improve generalization to new intents in a few-shot setting. Slot information (as a set of keywords) could still be used, to maximize consistency in the generated responses [24]. Furthermore, by applying the inherent world knowledge that these models are known to absorb, they could potentially detect out-of-scope utterances and respond appropriately. And, as *other modalities are worth a thousand words*, models which embed state in such a dynamic way could accept **image and video inputs**, which could also be jointly encoded into a general state representation.

To augment the real world knowledge these large-scale models possess, we could once again explicitly introduce relevant knowledge (from some knowledge base) to the model input. When training using well-known conversational assistant data, we can also use the slot-filling ground-truth to build a smaller, domain-appropriate, knowledge base. Successfully connecting NLU and the rest of the dialogue pipeline, either by jointly training or composing a single, larger, model, could help solve **the second challenge**.

5 CONCLUSION

Recently, I've had the opportunity to compete in the *competition*, and had my work interact with thousands of users. Observing their behaviors when interacting with a conversational assistant has given me some insight, and exacerbated some of my concerns. I wonder how far extremely large models can go until the gains stop being appealing. I think about data quality, and how we can analyze and use it in order to build something scalable and that generalizes well. And I wonder how these systems can be put in production and be continuously updated, while maintaining performance without expensive—and oftentimes ineffective—data collection. These systems are no longer simply command receptors. They interact with users in useful, natural and impactful ways. And it all begins by understanding what is being said.

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